

ABSTRACT

The “Amazon Product” offers an extensive range of products designed to meet the needs of diverse customers worldwide. Each product page serves as a gateway to detailed insights, ensuring users have all the information they need to make informed purchasing decisions. An Amazon product abstract provides a concise, engaging summary that captures the essence of the item, highlighting its key features and benefits in an accessible format. This abstract emphasizes what makes the product unique, how it addresses common user needs, and why it stands out from similar options.

The abstract generally focuses on the most important aspects of the product, such as its functionality, durability, innovative design, or ease of use, while weaving these details into a cohesive narrative. Whether the product is a piece of technology, a household essential, or a fashion item, the abstract highlights its practical applications, giving potential buyers a clear idea of how it can enhance their daily lives. Additionally, Amazon abstracts often incorporate relatable scenarios to demonstrate real- world value, helping customers visualize the product’s benefits.

By combining clear descriptions, persuasive language, and a focus on customer needs, an Amazon product abstract transforms essential information into a compelling story. This helps users quickly grasp the value of product, reinforcing their confidence in the purchase decision.

TABLE OF CONTENTS

CONTENTS	PAGE NO
ACKNOWLEDGEMENT	i
ABSTRACT	ii
TABLE OF CONTENTS	iii
LIST OF FIGURES	iv
Chapter 1: INTRODUCTION	1-5
1.1 Overview	1
1.2 Problem Statement	2
1.3 Objectives	2
1.4 Existing System	3
1.4.1 Disadvantages	3
1.5 Proposing System	4
1.5.1 Advantages	4
Chapter 2: LITERATURE SURVEY	6-7
Chapter 3: SYSTEM REQUIREMENT ANALYSIS	8-9
3.1 Hardware Requirements	8
3.2 Software Requirements	8
3.3 Functional Requirements	9
Chapter 4: SYSTEM ARCHITECTURE	10-15
4.1 Interaction Flow	10
4.2 Overview of System Architecture	13
Chapter 5: IMPLEMENTATION	16-26
5.1 Source Code	16
5.2 Flowchart	24
Chapter 6: RESULT AND SNAPSHOTS	27-28
CONCLUSION	29
FUTURE ENHANCEMENT	30
REFERENCES	31

LIST OF FIGURES

Figure No.	Figure Name	Page No.
4.1.1	Interaction Flow	12
4.2.1	Overview of System Architecture	14
5.2.1	Flowchart of Amazon product System	24
6.1	Folder Structure	27
6.2	Login Page	27
6.2	Home Page	38

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

A "Amazon Product" is an essential tool designed to systematically monitor and evaluate students' academic performance throughout their educational journey. This tool fosters a data-driven approach to education, enabling educators, students, and parents to have a clear and comprehensive view of academic progress. By aggregating metrics such as grades, attendance, assignment completion, test scores, and class participation, it provides a holistic view of a student's achievements and areas for improvement.

The tracker collects and analyzes data, representing each student with attributes like their name and a list of grades. This information is easily updated and retrieved, maintaining accurate records. By computing average grades, the tracker identifies strengths and weaknesses, facilitating targeted interventions to enhance learning outcomes.

Beyond individual performance, the tracker monitors overall course performance. By calculating average grades across all students, educators can gauge the academic health of their classroom, identifying trends and addressing potential issues early. This data is invaluable for curriculum planning and instructional adjustments. Interactive features include adding new students, recording and updating grades, and viewing detailed performance reports, streamlining administrative tasks.

Additionally, viewing the highest and lowest grades can highlight achievements and pinpoint students needing support. Transparent academic records bridge communication gaps among educators, students, and parents, fostering a collaborative environment for student success.

1.2 PROBLEM STATEMENT

In today's competitive online marketplace, consumers seek products that are reliable, efficient, and meet their specific needs. However, they often face challenges such as incomplete product information, unclear specifications, or a lack of real-world use cases. This gap can lead to confusion, uncertainty, and dissatisfaction with the purchasing experience. Additionally, buyers may be concerned about product durability, ease of use, compatibility, or whether the product truly delivers on its promises.

1.3 OBJECTIVES

The "Amazon Product" is a powerful tool that not only monitors and tracks academic performance but also provides a detailed and dynamic view of each student's progress. By consolidating data such as grades, attendance, assignment completion, and test results, it allows educators to see a holistic picture of student performance over time. This comprehensive data enables educators to detect trends, identify students who may need additional support, and implement targeted interventions that are tailored to the needs of each student.

The tracker promotes accountability by giving students the ability to regularly review their performance, track improvements, and set realistic academic goals. This fosters a sense of ownership in their learning journey and motivates them to stay engaged with their studies. Additionally, the system enhances communication by offering real-time access to performance records, allowing students, parents, and teachers to collaborate effectively in identifying areas for growth and ensuring that necessary support is provided.

Furthermore, the tracker allows educators to optimize teaching strategies by analyzing trends and performance metrics across different students and courses, helping to refine curriculum pacing, address gaps in understanding, and adapt teaching methods to various

learning styles. The automation of administrative tasks like grade entry, calculation, and reporting also reduces time spent on paperwork, enabling teachers to focus more on direct student engagement. In the long run, this system cultivates a more efficient and responsive educational environment, ensuring that students receive the support and guidance they need to achieve academic excellence.

1.4 EXISTING SYTEM

The existing system of a student academic progress tracker typically involves manual or semi-automated methods to record and monitor student performance. Educators maintain records of grades, attendance, assignments, and test scores, often using spreadsheets or basic software tools. These systems may lack integration, making it difficult to provide a comprehensive view of a student's progress. Data analysis is often time-consuming and prone to errors, limiting the ability to identify trends or implement targeted interventions effectively. Communication between educators, students, and parents is often hindered by fragmented or inaccessible records, reducing collaboration and transparency. Overall, traditional systems are functional but inefficient, highlighting the need for more streamlined, data-driven solutions.

1.4.1 DISADVANTAGES

- **Product Authenticity Issues:**

limit its effectiveness. One major issue is fragmented data, as academic records are often spread across different platforms or maintained manually.

- **Information Overloaded:**

This also leads to time-consuming processes as data entry and analysis require significant effort, delaying insights. Additionally, the system is prone to human errors, which can result in inaccurate performance assessments.

- **Environmental Impact:**

Traditional systems also lack advanced data analysis tools, limiting the ability to identify trends or address individual student needs effectively.

- **Return And Refund Challenges:**

Communication between educators, students, and parents is often hindered due to poor accessibility, with students and parents having limited access to real-time academic records.

1.5 PROPOSING SYSTEM

The proposed system is a web-based application designed to manage and track the academic progress of students within educational institutions. The system is built using Flask, a lightweight Python web framework, and offers a centralized platform for managing courses, students, and grades.

The system allows educators to create and manage courses, adding students to these courses and tracking their academic performance. Students can be added with their names and emails, and educators can assign grades, update student information, and remove students as necessary. Each student's grades are tracked over time, and educators can calculate both individual student averages and overall course averages, providing a clear picture of academic performance. The system also ensures that only authenticated users (such as administrators or teachers) can access certain features, utilizing login functionality to maintain data security. A simple authentication mechanism allows users to log in and out, with a session-based system that keeps them logged in until they log out manually.

The application features an intuitive interface where administrators can view and manage course details, individual student profiles, and the overall performance of students across all courses. Additionally, the system supports the ability to update student information and add grades dynamically, helping teachers maintain accurate and up-to-date academic records.

With its centralized, user-friendly interface, the system aims to streamline the management of student data, reducing administrative overhead, enhancing communication between educators and students, and supporting data-driven decision-making to improve academic outcomes. The system's core objectives are to improve academic tracking, facilitate better educational planning, and provide a collaborative environment for teachers, students, and parents.

1.5.1 ADVANTAGES

The "Amazon Product" offers several key advantages that contribute to its effectiveness in monitoring and managing student academic performance:

- 1. Wide Product Selection:** The tracker efficiently collects, organizes, and manages student data, including names and grades. This comprehensive approach ensures that educators have access to accurate and up-to-date information essential for making informed decisions about student progress.
- 2. Convenient Shopping Experience:** By maintaining records for each student, the enables educators to monitor individual academic performance over time. This granular

level of tracking allows for personalized interventions and support tailored to each student's needs.

- 3. Fast And Reliable:** In addition to tracking individual student performance, the tracker facilitates course-level analysis by calculating average grades for entire courses. Educators can identify trends, assess overall class performance, and make data-driven adjustments to instructional strategies as needed.
- 4. User-Friendly Interface:** The tracker features a user-friendly interface with menu-driven options that simplify the process of adding students, recording grades, and accessing performance data. This intuitive design enhances usability and promotes efficient use of the tool by educators and administrators.
- 5. Transparency and Collaboration:** By providing access to academic records and performance metrics, the tracker fosters transparency and collaboration among educators, students, and parents. Open communication regarding student progress enables stakeholders to work together effectively in supporting student success.
- 6. Real-Time Updates:** With its ability to update and retrieve data in real-time, the tracker ensures that educators have access to the most current information about student performance. This timely feedback allows for prompt intervention and support when needed, contributing to improved student outcomes.
- 7. Flexible and Scalable:** The tracker's modular design allows for flexibility and scalability, making it adaptable to various educational settings and student populations. Whether used in small classrooms or large institutions, the tracker can accommodate diverse needs and requirements.
- 8. Data-Driven Decision Making:** Educators can make informed decisions based on real-time data about student performance. This data-driven approach helps in tailoring teaching strategies to meet individual student needs.
- 9. Identifying Trends:** Educators can identify trends in student performance, such as common areas of difficulty or exceptional achievements, allowing for targeted instructional adjustments and interventions.

CHAPTER 2

LITERATURE SURVEY

[1] Andreas Papadakis, et al.,: “Profiling Students’ Performance and Measuring their Progress in the area of Multimedia Communications”

In this paper, a system for profiling students' progress in engineering disciplines is presented and built. This method allows students' status and advances in the field of communication systems to be evaluated and measured in a systematic manner. The framework is based on current engineering simulation tools that are commonly used in higher education.

[2] Anand Desai, et al.,: “Student Profiling to improve teaching and learning: A Data Mining Approach”

This study focuses on how data mining can be used in an educational environment. This study can be utilized to help schools classify students' student achievements as well as their perseverance as determined by a toughness assessment so that they can tailor their instruction to different groups of kids. For the required students, remedial lessons or extra examinations might be scheduled based on this classification. Students can also use the application to track their progress from semester to semester.

[3] Rafiqul Zaman Khan, et al.,: “A Study of Cloud Computing”

This study is based on the necessity for cloud computing, cloud computing architecture, cloud computing kinds, cloud computing services, and cloud computing offerings.

[4] Katsaros, Gregory, et al.,: “A Holistic View of Information Management in Cloud Environments”

Cloud computing gradually impacted the application computing environment, introducing new roles and responsibilities, as well as new systematic and business models. The amount of data available and the need to handle it properly grew in tandem with the increase of service providers' capabilities and potentials. To that end, we present in this article an administration service architecture as well as the data model upon which the result is based. The concept was created to allow a storage service to be offered by a Cloud computing platform, but it is adaptive enough to work in a variety of Cloud scenarios.

[5] Mehdi, et al.,: “IEEE International Conference on Mobile Cloud Computing”

The lesson will begin with an overview of cloud computing environments, cloud services design, the need for mobile or native cloud computing in the app industry to deal with new mobile development, internet tools, application creating tools, and the rationale for moving apps to cloud computing services. The course will cover the basics, objectives, and typical topologies of mobile cloud computing systems, as well as provide an introduction to generic mobile cloud services for app developers and marketers. This lesson will go over some of the most common issues and expenses, as well as the role of mobile cloud computing infrastructure in the field of app design and how the app industry may make a low cost transition to cloud computing systems.

**[6] “Systemic Risks in the Cloud Computing Model: Complex Systems Perspective”
by Marbukh, V**

The goal of this research is to quantify and manage the underlying systematic performance/risk tradeoff in the "cloud computing paradigm". The cloud is regarded as a complex system, and systemic hazards are associated with the likelihood of a framework phase transformation to an undesirable permanent state. The findings point to a change in cloud infrastructure development and management paradigms away from maximizing financial prosperity and toward managing and optimizing the underlying condition benefit/risk tradeoffs, which can be done using mean-field and fluid approximations. Economic factors are thought to intensify this tradeoff by pushing "Cloud service providers" closer to the operating regime's limit, increasing the risk of overload when the system's capacity is insufficient to fulfill exogenous demand.

[7] Yang, et al.,: “An Efficient and Secure Dynamic Auditing Protocol for Data Storage in Cloud Computing”

Data owners store their data on cloud storage, which users (data users) can gain entry through cloud computing technology. However, data exporting creates new security issues, necessitating the usage of a 3rd party monitoring service to assure integrity of data in the cloud. Because data in the cloud can be dynamically updated, several established faraway integrity checking methodologies can only be applied to passive archive material and thus cannot be used to audit the service.

CHAPTER 3

SYSTEM REQUIREMENT ANALYSIS

A system requirement analysis for Customer based Amazon product would typically involve identifying the hardware, software, and network requirements to ensure the effective functioning of the application. Here's an overview:

3.1 HARDWARE REQUIREMENTS:

- **CPU:** A modern single-core processor is sufficient for this console application. However, a dual-core or higher processor is recommended for better performance and to handle additional tasks or multitasking.
- **RAM:** At least 1 GB of RAM is adequate to run this Java program along with the necessary operating system and background processes. More RAM, such as 2 GB or higher, will ensure smoother performance, especially if running multiple applications simultaneously.
- **Storage:** At least 100 MB of free disk space for the Python environment and necessary packages. Additional space may be required for more complex applications or datasets.
- **Display:** A monitor with a resolution of 1024x768 or higher is recommended for comfortable viewing of the console output.

3.2 SOFTWARE REQUIREMENTS:

- **html:** Version 3.6 or higher.
- **Flask:** It is a lightweight web framework used to build the application. It handles the HTTP routes, request/response cycle, rendering templates, and managing sessions.
- **Web Browser:** Any modern browser such as Google Chrome, Firefox, Microsoft Edge, or Safari for accessing the web interface.
- **Text Editor/IDE:** A code editor like Visual Studio Code, PyCharm, Sublime Text, or any other preferred editor for editing Python files.
- **Operating System:** Compatible with Windows, macOS, or Linux systems.
- **Command-Line Interface (CLI) / Terminal:** To run the Flask application, you will use a terminal or command-line interface to navigate to the directory containing app.py and start the server with the command `python app.py`.

3.3 FUNCTIONAL REQUIREMENTS:

- **Input Handling:** The program requires an input mechanism to receive user input for various operations such as adding students, adding grades, calculating averages, viewing data, etc. This input mechanism can be through the console using a keyboard or any other input device.
- **User Interface:** Although the code doesn't specify a graphical user interface (GUI), it requires a text-based interface to display prompts, menus, and messages to the user. This can be achieved through a console or terminal window.
- **Storage:** The program needs a method to store and manage data about students and their grades. It currently utilizes in-memory data structures such as lists for storing student and grade information. However, for a more robust solution, a database or file storage system could be implemented to persist data between program executions.
- **Student and Course Management:** The system should support functions to add, remove, and update student and course information. This includes adding students to courses, adding grades for students, removing students from courses, etc.
- **Calculation Functions:** The program contains several calculation functions, such as calculating the average grade for a student or the entire course, finding the highest and lowest grades, etc. These functions must accurately compute the required values based on the provided data.
- **Error Handling:** Functional requirements include error handling mechanisms to manage unexpected inputs or situations. For example, the program should handle cases where a student or course is not found, incorrect grade inputs, invalid menu choices, etc.
- **Menu System:** The system requires a menu system to allow users to navigate and perform various operations. The menu should provide clear options for each operation and guide the user through the available functionalities.
- **Initialization:** At the start of execution, the program should initialize any necessary components, such as creating a course object and displaying a welcome message to user.
- **Looping Mechanism:** The program should implement a looping mechanism to continuously prompt the user for input and execute the selected operations until the user chooses to exit the program.
- **Clean Exit:** Finally, the program should provide a clean and user-friendly exit mechanism, ensuring that any resources are properly released, and the program terminates gracefully when the user decides to exit.

CHAPTER 4

SYSTEM ARCHITECTURE

4.1 INTERACTION FLOW

1. Login:

- **Action:** The user visits the /login route.
- **If Request is GET:** The login.html page is rendered with a login form.
- **If Request is POST:**
 - The application checks the credentials (username == "admin" and password == "password").
 - **Successful Login:** A session key logged_in is set to True, and the user is redirected to the / (index) route with a success flash message.
 - **Invalid Credentials:** A failure message is flashed, and the login form is re-rendered.

2. Home Page (Index):

- **Action:** After successful login, the user lands on the / route (the home page), which shows the list of courses.
- **Restricted to logged-in users:** The login_required decorator ensures only authenticated users can access the home page.
- **Actionable Items:**
 - Add a course via a form.
 - View courses and manage students.

3. Course Management:

- **Adding a Course:**
 - **Action:** A POST request to /add_course with the course name.
 - The application creates a new Course object and appends it to the courses list.
 - The user is redirected to the home page with a flash message confirming the course has been added.

- **Course Details:**

- **Action:** The user clicks on a specific course name to go to /course/<course_name>.
- The page shows all students enrolled in that course along with an option to add/remove/update students or add grades.

4. product Management:

- **Adding a Student:**

- **Action:** A POST request to /add_student/<course_name> with the student's name and optional email.
- The student is added to the respective course.
- The user is redirected to that course's details page with a flash message confirming the addition.

- **Removing a Student:**

- **Action:** A POST request to /delete_student/<course_name> with the student's name.
- The student is removed from the respective course, and the user is redirected to the course details page with a flash message confirming the removal.

- **Updating Student Information:**

- **Action:** A POST request to /update_student/<course_name> with the old name, new name, and optional email.
- The student's information is updated, and the user is redirected with a flash message confirming the update.

- **Product Details:**

- **Action:** The user clicks on a student's name to go to /product/<product_name>/<customer_name>.
- The page displays detailed information for the selected student, including grades.

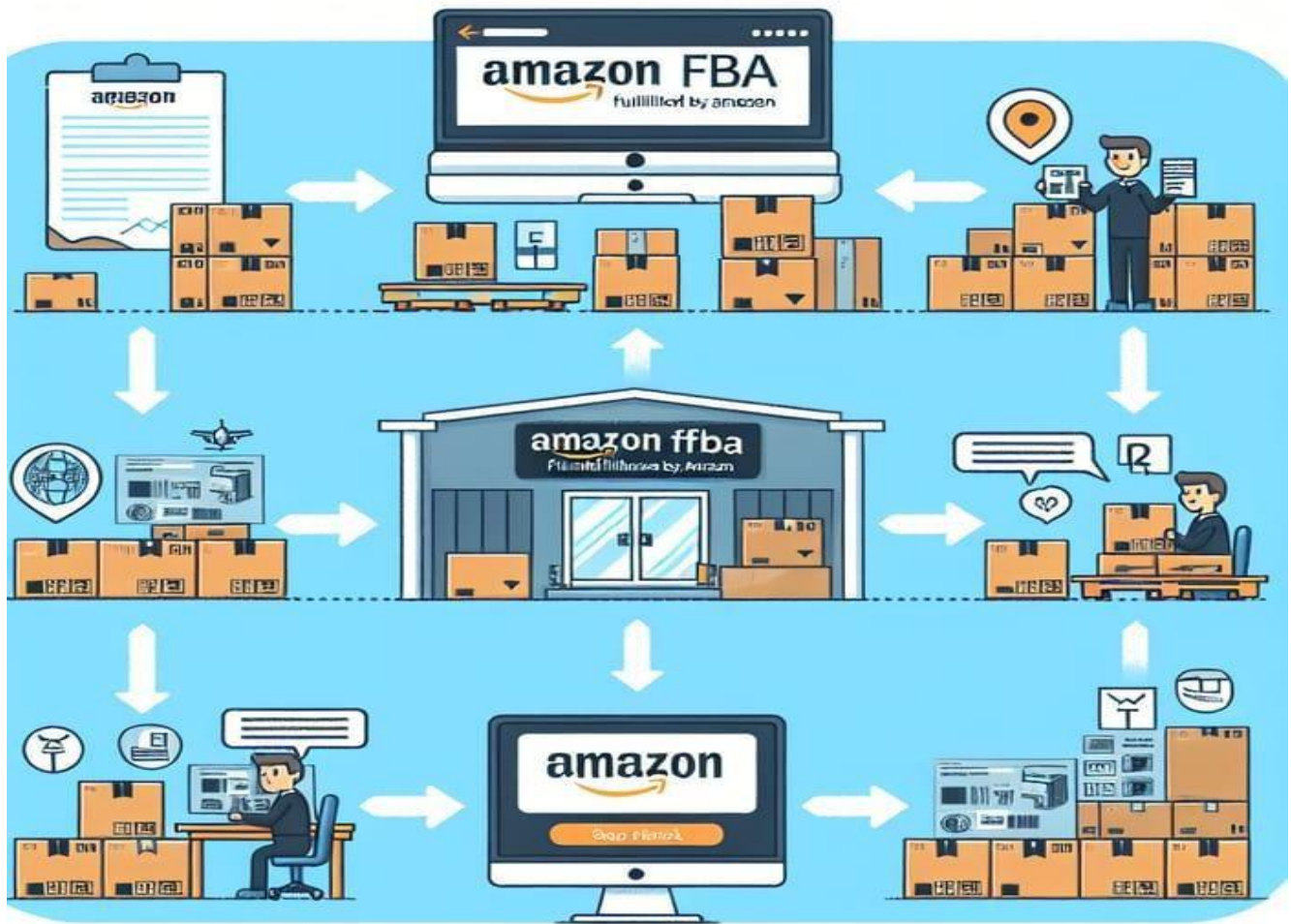


Fig 4.1.1: Interaction Flow

5. Product Management:

- **Adding Review:**

- **Action:** A POST request to `/add_grade/<course_name>` with the student's name and grade.
- The grade is added to the student's record, and the user is redirected with a flash message confirming the addition.

6. Average Customer Review:

- **Calculating Customer Average:**

- **Action:** A request to `/calculate_average_course_grade/<course_name>`.
- The application calculates the average grade for the entire course (using each student's average grade) and renders the `average_course_grade.html` template displaying the result.

- The page displays a list of all students across all courses.

7. Logout:

- **Action:** A GET request to /logout.
- The session is cleared (logged_out), and the user is redirected to the login page with a flash message confirming successful logout.

Key Points in Interaction Flow:

- **Authentication:** The app requires users to log in before performing any operations.
- **Customer and Product:** Users can create courses, add students, update student details, remove students, and add grades.
- **Review and Averages:** Users can manage grades for students and view average grades at both the student and course levels.
- **Flash Messages:** For user interaction feedback, all actions such as successful login, adding/removing students, updating, and adding grades are followed by flash messages, keeping the user informed.

Theoretical Flow Summary:

- **Login:** User logs in → Home page with courses → Add/Remove courses, students, and grades → View student details → Log out.

This process ensures a controlled and authenticated environment for managing courses, students, and their grades.

4.2 OVERVIEW OF SYSTEM ARCHITECTURE

1. Presentation Layer:

➤ Frontend (HTML Templates):

- Uses Jinja2 templates to render pages dynamically.
- Templates include forms and pages such as login, index, course details, and student details.

➤ Forms:

- User inputs for login, adding/updating courses, students, and grades are collected using HTML forms.

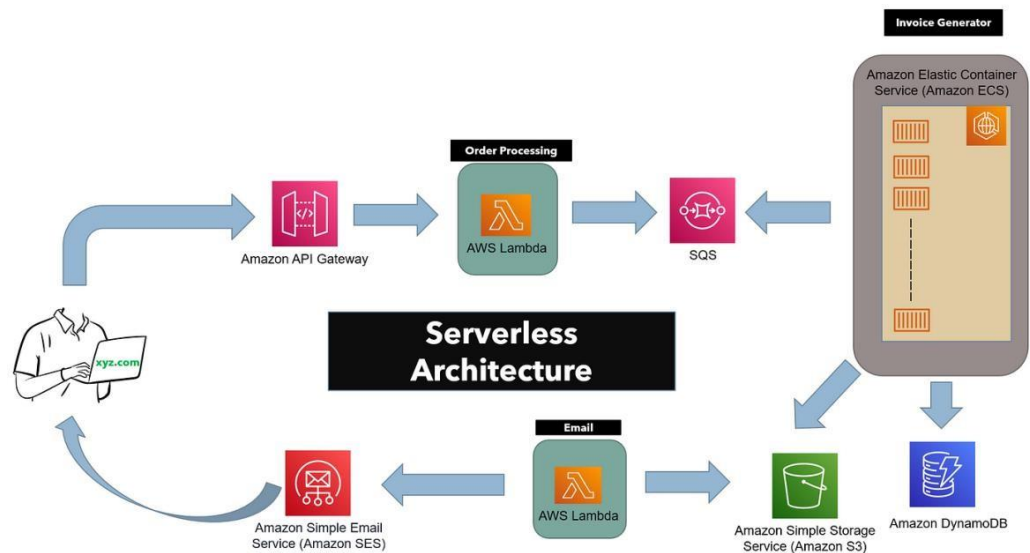


Fig 4.2.1: Overview of System Architecture

2. Application Layer:

➤ Flask Application:

- Handles HTTP requests and responses using Flask's routing mechanism.
- Contains view functions (routes) for each interaction in the application (e.g., adding students, managing courses).
- Implements session management to control user authentication and access.

3. Business Logic Layer:

➤ Customer care and Course Classes:

- Manage student-related operations such as adding grades, updating information, and calculating averages.
- Handle course-level operations, including adding/removing students and calculating average grades.

4. Data Management:

➤ In-memory Data Storage:

- The current implementation uses in-memory lists to store courses and students (not persistent between runs).

➤ Potential for Database Integration:

- The application can be extended to use databases such as SQLite, PostgreSQL, etc., for data persistence.

5. Security:

➤ Session Management:

- Implements a simple session-based authentication system using Flask sessions and a secret key.

➤ Form Handling:

- Basic validation and flash messages for use feedback.

The system architecture of the Product based Amazon Product based follows a client-server model, where the client interacts with the application through a web interface, and the server handles data processing and management. At the core of the system is the Flask web framework, which serves as the server-side component. The server processes incoming HTTP requests, handles user authentication, manages student and course data, and performs necessary computations like grade averaging. It relies on in-memory data structures, such as lists of Course and Student objects, to store and manipulate information. The client-side interacts with the system through HTML templates rendered by Flask's Jinja2 template engine, allowing users to add courses, manage students, assign grades, and view performance statistics. The system also uses Flash messages to provide real-time feedback on user actions. Authentication is handled via session management, ensuring that only authorized users (e.g., administrators) can perform sensitive actions like adding or removing students. The modular structure of the system allows for easy expansion, such as integrating a persistent database for data storage or adding more complex features for data analysis. This architecture provides a robust and efficient platform for managing student progress while ensuring a user-friendly experience through dynamic web pages.

CHAPTER 5

IMPLEMENTATION

5.1 SOURCE CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Amazon Product Showcase</title>
  <!-- Bootstrap CSS -->
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0-alpha1/dist/css/bootstrap.min.css" rel="stylesheet">
  <style>
    body {
      background-color: #f9f9f9;
    }
    .product-card
      img{ height:
      200px; object-
      fit: cover;
    }
    .product-card {
      box-shadow: 0 4px 8px rgba(0, 0, 0,
      0.1); border-radius: 8px;
      margin-bottom: 20px;
    }
    .product-card .btn {
      background-color:
      #FF9900; color: white;
    }
    .product-card .btn:hover {
      background-color:
      #cc7a00;
    }
    footer {
      background-color:
      #343a40; color: #fff;
      text-align: center;
      padding: 20px 0;
    }
    header {
```

```

        background: #f8f9fa;
        padding: 50px 0;
text-align: center;
    }
    header h1 {
        font-size: 2.5rem;
    }
    nav a {
        margin-right:
        15px; color:
        #000;
    }
    nav a:hover {
        color:
        #FF9900;
    }
    #product-container .col-md-
        4 { margin-bottom: 30px;
    }
</style>
</head>
<body>
    <!-- Header -->
    <header>
        <div class="container">
            <h1>Welcome to Amazon</h1>
            <p>Your one-stop shop for amazing products at unbeatable prices!</p>
            <a href="#products" class="btn btn-primary">Shop Now</a>
        </div>
    </header>

    <!-- Navbar -->
    <nav class="navbar navbar-expand-lg navbar-light bg-light">
        <div class="container">
            <a class="navbar-brand" href="#">Amazon</a>
            <button class="navbar-toggler" type="button" data-bs-toggle="collapse"
data-bs- target="#navbarNav">
                <span class="navbar-toggler-icon"></span>
            </button>
            <div class="collapse navbar-collapse" id="navbarNav">
                <ul class="navbar-nav ms-auto">

```

```

text-align: center;
}
header h1 {
  font-size: 2.5rem;
}
nav a {
  margin-right:
  15px; color:
  #000;
}
nav a:hover {
  color:
  #FF9900;
}
#product-container .col-md-
4 { margin-bottom: 30px;
}
</style>
</head>
<body>
<!-- Header -->
<header>
  <div class="container">
    <h1>Welcome to Amazon</h1>
    <p>Your one-stop shop for amazing products at unbeatable prices!</p>
    <a href="#products" class="btn btn-primary">Shop Now</a>
  </div>
</header>

<!-- Navbar -->
<nav class="navbar navbar-expand-lg navbar-light bg-light">
  <div class="container">
    <a class="navbar-brand" href="#">Amazon</a>
    <button class="navbar-toggler" type="button" data-bs-toggle="collapse"
data-bs- target="#navbarNav">
      <span class="navbar-toggler-icon"></span>
    </button>
    <div class="collapse navbar-collapse" id="navbarNav">
      <ul class="navbar-nav ms-auto">
        <li class="nav-item"><a class="nav-link" href="#">Home</a></li>
        <li class="nav-item"><a class="nav-link" href="#products">Products</a></li>

        <li class="nav-item"><a class="nav-link" href="#about">About</a></li>

```

```

        <li class="nav-item"><a class="nav-link" href="#contact">Contact</a></li>
    </ul>
</div>
</div>
</nav>
    <a href="https://www.amazon.in/boAt-Rockerz-430-Assistant-Headphones/dp/B0DGTSTQWL/ref=sr_1_1_sspa?crd=85FALHCBFAE0&dib=eyJ2IjojMSJ9.51DoRHVn-9oTkFmrLLR8VQ-qPq7FrAsbtV_VSOl0anNq1dk6gY4EnMOAnQHHLKTX6rdBvF_d0a-i2-GriKf4LWTPUoXCSAiKfwgSuDT5HjuIdNHfM-wipkNdIcWysIP2WjLgBdpPysAMVuTrCSglorIMYrEyUzSg1cpTLfnTGxhknXUQ9hEa9ymLAte0_YjEYadGkxhdNFuprDIggCdQ1fXqGVPRcph7ZHXiO0w7FYw._wt3BIHm2LjZQwCtCYyvel0tHLrKkf dyVuhs B8-24qA&dib_tag=se&keywords=wireless%2Bheadphones%2Bboat&nsdOptOutParam=true&qid=1732592694&sprefix=wireless%2Bhead%2Caps%2C264&sr=8-1-spons&sp_csd=d2lkZ2V0TmFtZT1zcF9hdGY&th=1" target="_blank" class="btn btn-block">Read More</a>
</div>
</div>
</div>
<div class="col-md-4">
    <div class="card product-card">
        
        <div class="card-body">
            <h5 class="card-title">Laptop</h5>
            <p class="card-text">High-performance laptop perfect for work and entertainment.</p>
            <p class="text-muted">₹99999</p>
            <a href="https://www.google.com/url?q=https://www.vijaysales.com/microsoft-surface-laptop-go-3-12th-gen-core-i5-8-gb-ram-256-gb-ssd-12-4-inch-31-49cm-display-intel-iris-xe-graphics-win-11-xk1-00045/25484&opi=95576897&sa=U&ved=0ahUKEwjUh_j9xPCJAxWkgVYBHfl6D5UQsDwI6Qg&usg=AOvVaw1DLWZQ_dWW3y6LdfP89xP6" target="_blank" class="btn btn-block">Read More</a>
        </div>
    </div>
</div>
</div>
<!-- More Products -->
<div class="col-md-4">
    <div class="card product-card">
        
        <div class="card-body">
            <h5 class="card-title">Tablet</h5>
            <p class="card-text">Compact and powerful, great for browsing and gaming on the go.</p>

```

```
<p class="text-muted">₹10000</p>
<a href="https://www.amazon.in/Samsung-Galaxy-Display-Expandable-
Tablet/dp/B0CHZ38BPP/ref=sr_1_1_sspa?crid=YUGH6K2YVXDG&dib=eyJ2IjoMSJ9.oVUYK
PY5iYA ygAINro2dWn7btRYHWsCjA-
wYwjMTW4kofKdSNJtVSctztFSnEM0xGJ2UsUaTqm9i6ZD4Tf8TIA-
qMwlV4zPb8hoZICijRcsPnCd7RRpckbwo5HGavLWt7TOBOsVS6G5SX_zrEOXjLtDhMHmw
Q- MW7HpjGADvDLUoVmmHEenU_EKGGq9M3VThCPK3ANmqW3dm-paHll-
LZt1pPzm0WvwWvq27JdXeZNo.N79M-
ofCMFjoCeFoVwCVNkzoJfdqzvtdCDLqP5iyNBQ&dib_tag=se&keywords=tabs+5g&qid=173259
2637&s prefix=tabs%2Caps%2C262&sr=8-1-
spons&sp_csd=d2lkZ2V0TmFtZT1zcF9hdGY&psc=1" target="_blank" class="btn btn-
block">Read More</a>
</div>
</div>
</div>
<div class="col-md-4">
  <div class="card product-card">
    
    <div class="card-body">
      <h5 class="card-title">Gaming Console</h5>
      <p class="card-text">Immersive gaming experience with high-
performance capabilities.</p>
</div>
</div>
<div class="col-md-4">
  <div class="card product-card">
    
    <div class="card-body">
      <h5 class="card-title">Laptop</h5>
      <p class="card-text">High-performance laptop perfect for work and entertainment.</p>
      <p class="text-muted">₹99999</p>
      <a href="https://www.google.com/url?q=https://www.vijaysales.com/microsoft-
surface-laptop-go-3-12th-gen-core-i5-8-gb-ram-256-gb-ssd-12-4-inch-31-49cm-display-intel-iris-
xe-graphics-win-11-xk1-
00045/25484&opi=95576897&sa=U&ved=0ahUKEwjUh_j9xPCJAxWkgVYBHfl6D5UQsDwI6Qg
&usg= AOvVaw1DLWZQ_dWW3y6LdfP89xP6" target="_blank" class="btn btn-block">Read
More</a>
      <p class="text-muted">₹3999</p>
      <a href="https://www.amazon.com/example-gaming-console" target="_blank"
class="btn btn-block">Read More</a>
    </div>
  </div>
</div>
<div class="col-md-4">
```

```
<div class="card product-card">
  
  <div class="card-body">
    <h5 class="card-title">Camera</h5>
    <p class="card-text">Capture stunning photos with this high-quality camera.</p>
    <p class="text-muted">₹59999</p>
    <a href="https://www.amazon.in/Vlogging-Camera-4K-Digital-
YouTube/dp/B0BGLDXHGY/ref=sr_1_3?crd=3AQUS2RMY94LX&dib=eyJ2IjoiMSJ9.KuZs3dT
PDctlQ k3o4pTfBdJPxhDqNLkYnu9iQzekUGtJfzQjZERrFs0PkH-
```

```
QgjK0EyUixSbTm2iaMmzHXZsNdKWfFyVSof69uyDjKFty18m6aJzspjxRlSmiKxfJllkwcf2lZ4
pJ_xvfX
KA2gefovD4eqOu3GV0oNjkcno6u23QjtEw8dHjlrPon3B9GHOnDIA3hYau27JpLMZGj6hgRY
RjKoR3 O-
uASwyTICqx3R8.kmN8KQElxb24a8vFZ6IQ4PDYhXuGXpgMzmIUf9lfaoU&dib_tag=se&keyw
ords=camara+dslr&qid=1732592569&sprefix=cemara%2Caps%2C254&sr=8-3"
target="_blank" class="btn btn-block">Read More</a>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
<!-- About Section -->
```

```
<section id="about" class="bg-light py-5">
```

```
<div class="container text-center">
```

```
<h2>About Us</h2>
```

```
<p>We are dedicated to bringing you the best shopping experience with top-quality products.</p>
```

```
</div>
```

```
</section>
```

```
<!-- Contact Section -->
```

```
<section id="contact" class="py-5">
```

```
<div class="container">
```

```
<h2 class="text-center mb-4">Contact Us</h2>
```

```
<form>
```

```
<div class="mb-3">
```

```
<label for="name" class="form-label">Name</label>
```

```
<input type="text" class="form-control" id="name" placeholder="Your Name">
```

```
</div>
```


Resolution/dp/B0BF4YBLPX/ref=sr_1_1_ssapa?crd=1DYV8ZV6HGQH0&dib=eyJ2IjojMSJ9.gK4V6eYn
tJdXzMFBAAqam9vKuzvMbHtM_Em0iEluOlKbIOEwZWlSbhXR7bV0Xh_PHXYsq2H3t_wg5F
omOhhI- x8IfzPi-
z5Pg1EPFXbOADzQNFeHTsojdgf0hCyRuW83cywraAp39NKfc5rAu9S_QOxsuX6hRnRROQ8C
4kcGN3
nVhSDO8U7ssoCteZYwpms1EuL_raNYZ6WfK1LVNjTXBWfBmQYU64QNqAHWNTD2QxKg.
t3_Tje2
XyUlpy1feSQu3pNiTIx2A9K1mfRYzBYgufyY&dib_tag=se&keywords=smart+watch+for+wom
en+latest &qid=1732592781&sprefix=smart%2Caps%2C327&sr=8-1-
spons&sp_csd=d2lkZ2V0TmFtZT1zcF9hdGY&psc=1 " target="_blank" class="btn btn-
block">Read More

</div>

</div>

</div>

<div class="col-md-4">

<div class="card product-card">

<div class="card-body">

<h5 class="card-title">Wireless Headphones</h5>

<p class="card-text">Experience superior sound quality with these wireless
headphones.</p>

<p class="text-muted">₹89999</p>

<a href="https://www.amazon.in/boAt-Rockerz-430-Assistant-
Headphones/dp/B0DGTSTQWL/ref=sr_1_1_ssapa?crd=85FALHCBFAE0&dib=eyJ2IjojMSJ9.5
1DoRHV n-9oTkFmrLLR8VQ-

qPq7FrAsbtV_VSOl0anNq1dk6gY4EnMOAnQHHLKTX6rdBvF_d0a-i2-
GriKf4LWTPUoXCSAiKfwgSuDT5HjuIdNHfM-
wipkNdIcWysIP2WjLgBdpPysAMVuTrCSglorIMYrEyUzSg1cpTLfnTGxhknXUQ9hEa9ymLAte
0_YjEY
adGkxhdNFuprDIggCdQ1fXqGVPRcph7ZHXiO0w7FYw._wt3BlHm2LjZQwCtCYyvel0tHLrKkf
dyVuhs B8-

<!-- Contact Section -->

<section id="contact" class="py-5">

<div class="container">

<h2 class="text-center mb-4">Contact Us</h2>

<form>

<div class="mb-3">

```
<label for="name" class="form-label">Name</label>
```

```
<input type="text" class="form-control" id="name" placeholder="Your Name">
</div>
```

```
<!-- Contact Section -->
```

```
<section id="contact" class="py-5">
```

```
<div class="container">
```

```
<h2 class="text-center mb-4">Contact Us</h2>
```

```
<form>
```

```
<div class="mb-3">
```

```
<label for="name" class="form-label">Name</label>
```

```
<input type="text" class="form-control" id="name" placeholder="Your Name">
```

```
</div>
```

```
<div class="mb-3">
```

```
<label for="email" class="form-label">Email</label>
```

```
<input type="email" class="form-control" id="email" placeholder="Your Email">
```

```
</div>
```

```
<div class="mb-3">
```

```
<label for="message" class="form-label">Message</label>
```

```
<textarea class="form-control" id="message" rows="4" placeholder="Your
Message"></textarea>
```

```

```

```
<div class="card-body">
```

```
</div>
```

```
<button type="submit" class="btn btn-primary">Send Message</button>
```

```
</form>
```

```
</div>
```

```
</section>
```

```
<!-- Footer -->
```

```
<footer>
```

5.2 FLOWCHART

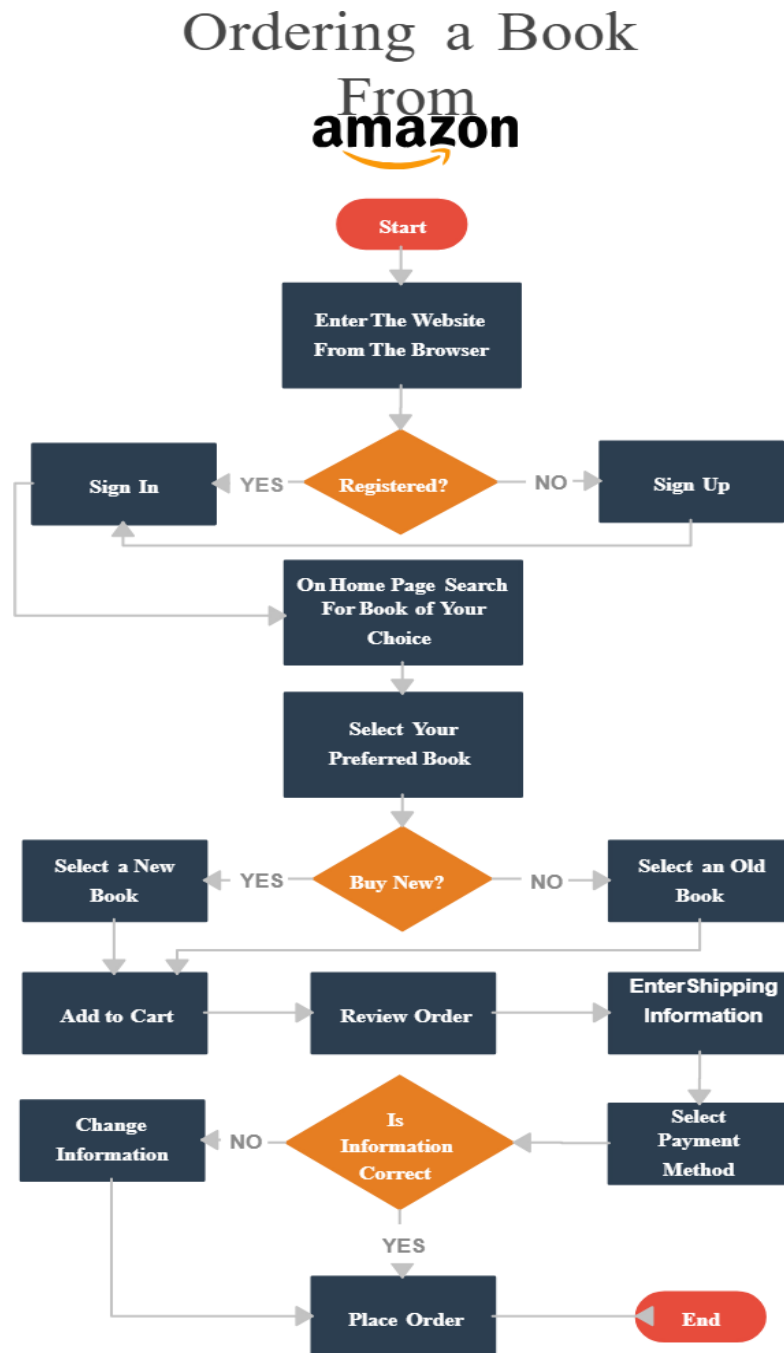


Fig 5.2.1: Flowchart of Amazon Product

The above fig 5.2.1 represents flowchart of a “**Amazon product based**” system, focusing specifically on the management of student check-in data, which is essential for monitoring attendance and academic engagement

The system offers three core functionalities. The first is **Product based Check-In Registration**, where administrators can record new check-ins by entering critical details such as student identification, check-in time, and the purpose of attendance. This feature is vital for maintaining accurate attendance records, a key component of any academic progress tracking system. The second functionality, **Modify Product Check-In Information**, allows for updates or corrections to existing records. For example, if an error is detected in a recorded check-in time or student ID, this feature ensures data accuracy by enabling modifications. Accurate data is essential for generating reliable reports on student attendance trends and engagement levels. The third functionality, **Delete Product Check-In Information**, provides a means to remove redundant or erroneous records. This ensures the database remains clean and relevant, supporting efficient analysis of attendance data.

Once all necessary tasks are completed, the administrator can exit the system through the **Quit Module**, securely ending their session. The structured workflow outlined in the flowchart ensures a systematic approach to managing attendance data, which is critical for tracking academic progress. By maintaining accurate attendance records, institutions can identify patterns of absenteeism, assess student engagement, and take proactive measures to support academic success. This system serves as an essential tool for administrators and educators in fostering a data-driven approach to improving student outcomes.

Additionally, the system provides a detailed review dashboard that displays real-time updates and summaries of student check-ins, allowing administrators to quickly view attendance patterns. The system's ability to track attendance by course or time slot helps pinpoint areas where students may be struggling due to low engagement or absenteeism.

The Delete Product Check-In Information feature prompts administrators to verify the record before deletion to avoid accidental errors. The system also integrates a customizable reporting feature, generating daily, weekly, or monthly attendance reports for individuals or entire classes.

Furthermore, the system integrates a reporting feature, which can generate daily, weekly, or monthly attendance reports. These reports are customizable and can be used for both individual students and entire classes. This functionality is crucial for monitoring overall customer participation, identifying students who need additional academic support, and informing decisions related to course scheduling or intervention programs.

By maintaining accurate customer review, the system not only supports academic tracking but also contributes to improving student engagement. This comprehensive approach helps educators, administrators, and students make more informed decisions, ultimately promoting academic excellence.

CHAPTER 6

RESULT AND SNAPSHOTS

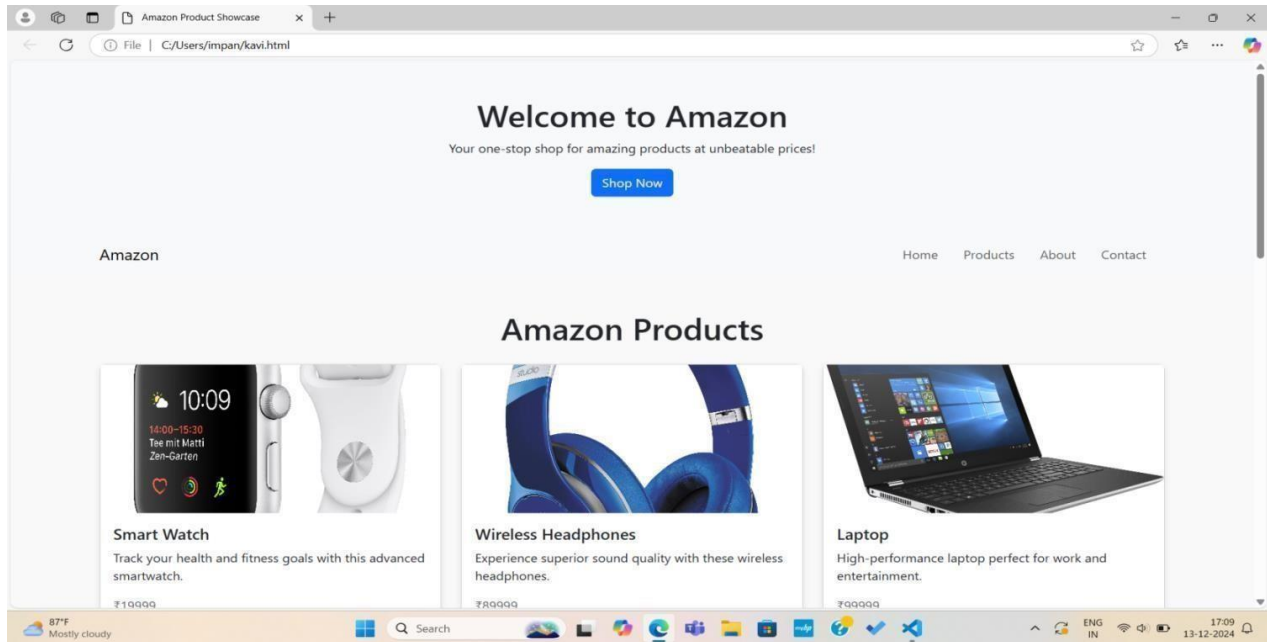


Fig 6.1: it describe the Amazon product homepage

The above figure 6.1 is a Folder Structure which is a way to organize and manage files and directories (folders) in a hierarchical layout on a computer or storage system. It determines how files and folders are grouped, named, and accessed, which helps in maintaining clarity,

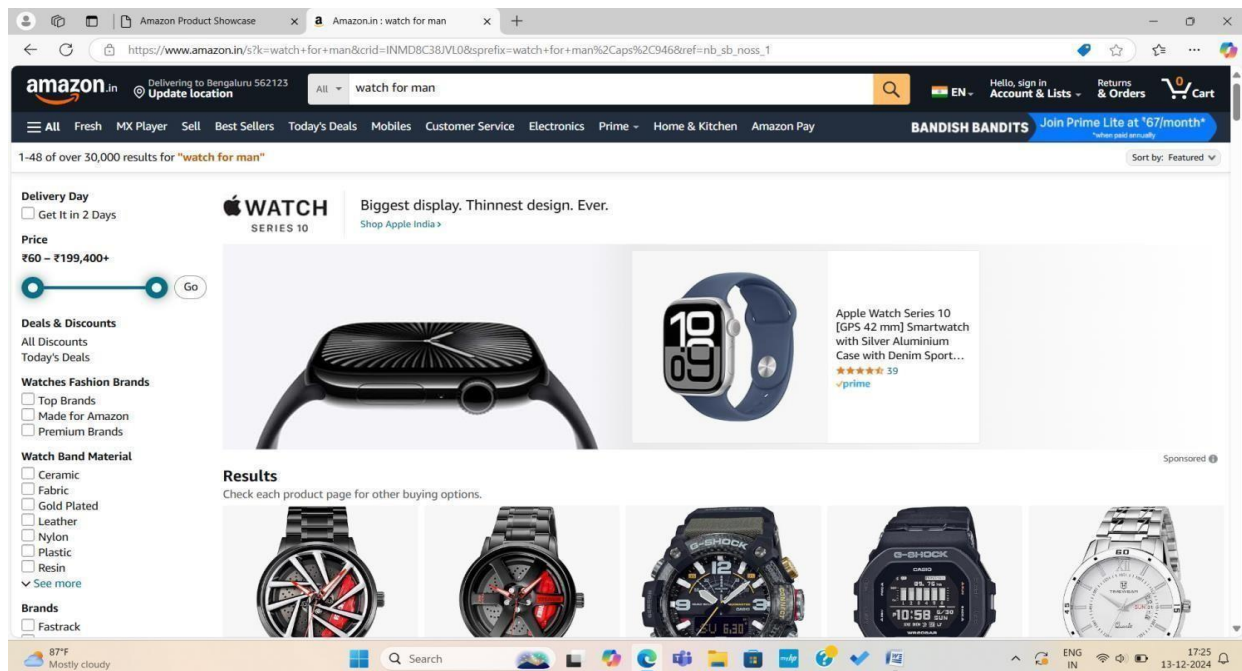


Fig 6.2: Login Page

The above figure 6.2 is a Login page. This is a first window appears when it is executed. It is an interface where users enter their credentials. The login page prompts users to enter their credentials (username and password). Upon successful login (with predefined credentials "admin" and "password"), the user is redirected to the home page with a success message. Invalid credentials prompt a flash message notifying the user to try again.

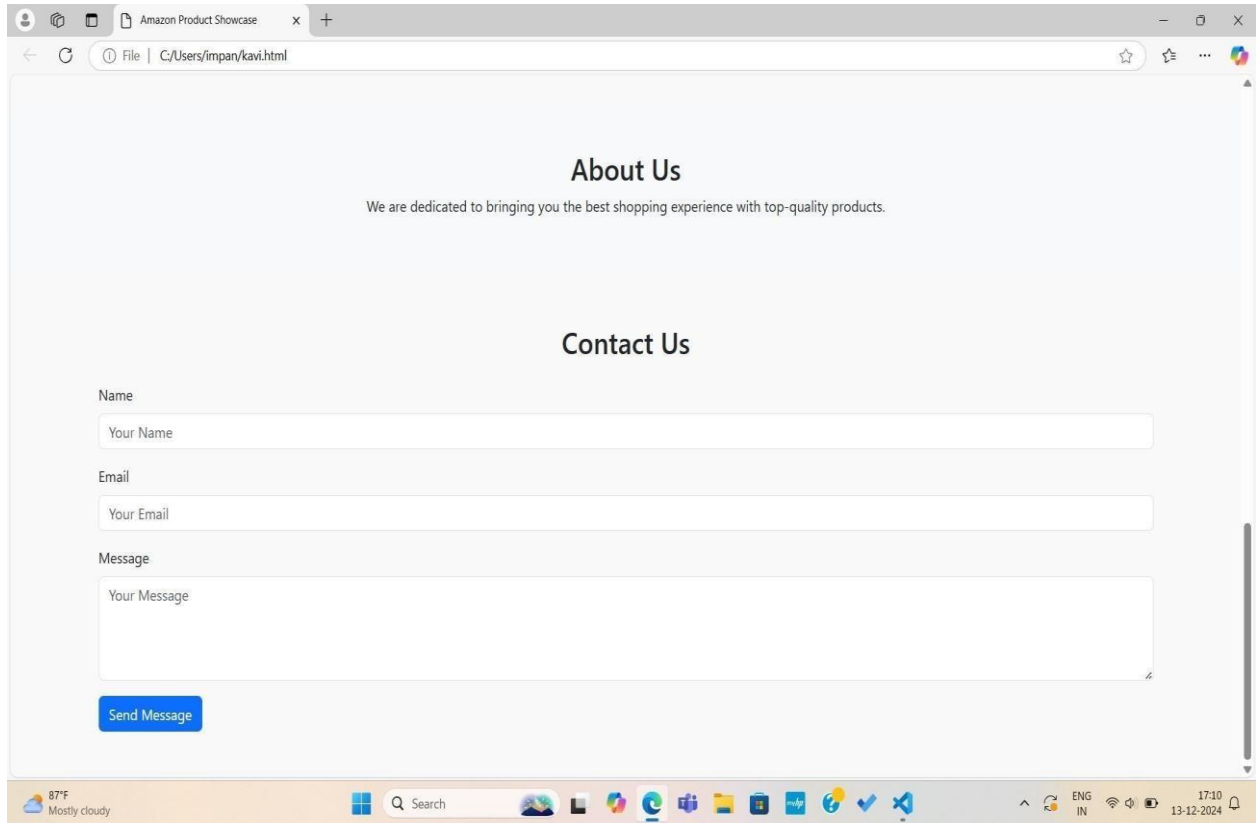


Fig 6.3: Home Page

The above figure 6.3 is a Home Page which is the main landing page of a website or application, serving as the central hub for navigation and showcasing key content or features. It often includes a menu, highlights important sections, and provides a starting point for users to explore further.

CONCLUSION

In conclusion, a Product based Progress Tracker stands as a powerful instrument in modern education, seamlessly blending technology with pedagogy to enhance people learning experiences. By harnessing data and analytics, this tool empowers educators with actionable insights to personalize instruction, identify struggling students, and implement targeted interventions. Moreover it fosters a collaborative ecosystem, enabling constructive dialogue between customer, product, and which by providing transparent and accessible academic records. Amazon has its task cut out as far as its future strategies are concerned and this SWOT Analysis can provide a guide and a roadmap that the company can implement going forward. The key take away from this SWOT Analysis is that Amazon has to focus on profitability and not volumes alone if it has to be competitive in the future where volumes and market leadership are not alone to add value to its stock. SWOT analysis clarifies the current standing of Amazon. Few necessary improvements are needed to be done to administer the lacking and reinforce its market position.

FUTURE ENHANCEMENT

The Future application can be enhanced by incorporating features for better functionality, security, and user experience. User roles can be introduced to allow admins to manage courses and students, while students only access their grades. Password hashing can replace plain-text passwords to increase security. To ensure data persistence, integrating a database like SQLite or PostgreSQL would be beneficial, with an ORM like SQLAlchemy to manage data instead of using in-memory lists. The grade management system can be improved by validating grade inputs and storing a history of grades for each student. Additionally, a student dashboard can provide a personalized view of their courses and grades, with profile management options to update personal details. Teachers could upload course content, grade assignments, and send email notifications. On the front-end, a responsive design using frameworks like Bootstrap or Tailwind CSS can improve the user interface, while interactive charts or graphs can help visualize student performance. Implementing AJAX for dynamic updates and providing detailed reporting features will enhance the overall experience. Robust error handling, logging, and unit tests are also essential for maintaining reliability and scalability of the application.